Design and Analysis of Software Systems
Final Exam – Spring 2025
29th April 2025

Name: ____________________________ **Invigilator Signature:**_________________

Roll number: _____________________ **Your signature:** ______________________

Question cum Answer Booklet !

INSTRUCTIONS: This is a closed-book, closed-notes exam. You may use a one page 8 ½ * 11 inch **hand-written** cheat sheet. Xerox copies or any downloaded notes/slides is not allowed.

It consists of 17 questions. You are to answer all questions. You have 180 minutes to complete the exam for a total of 80 points.

Credit is given for what you write, not what you are thinking. Partial credit will be given based on Quality of the content, not Quantity. A voluminous content-free answer may simply annoy the grader!!!

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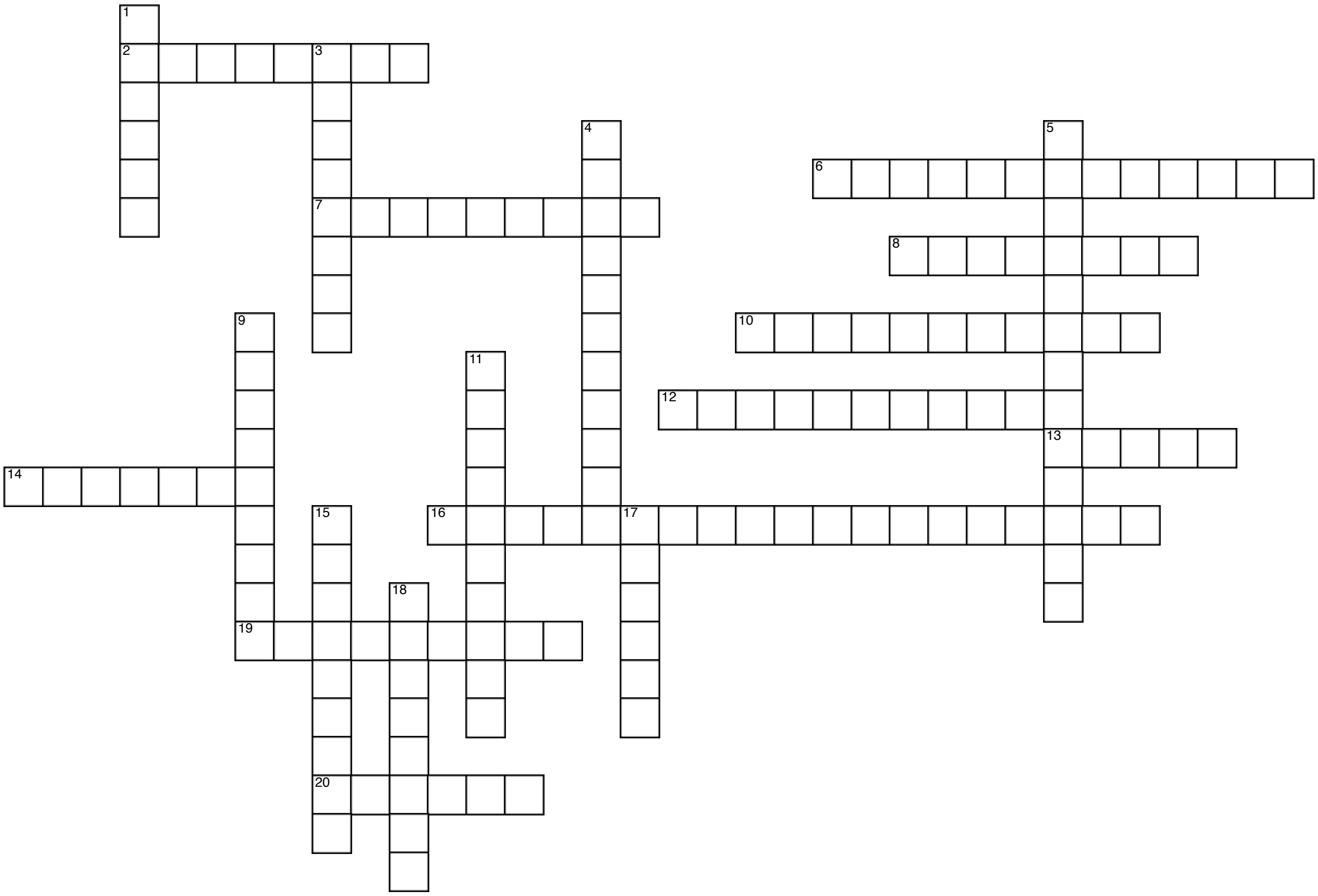
Q1. Please solve the crossword puzzle. (20 points)
Based on the clues provided below, identify the correct words that have been discussed during the lectures. Some answers may consist of two words; however, please write them as a single combined word without any spaces when filling in the puzzle. Keep in mind that while multiple answers may seem plausible, only one word is the correct fit.



ACROSS
2 Problem-solution pairs that have been discovered to be good over the years
6 Grouping your application into logical parts
7 There may be may _____ flows in a use case
8 process of restructuring existing computer code
10 This relates one class to another
12 this principle involves hiding complex realities with simpler interfaces
13 This is the initial phase of testing where developers test the product internally
14 refers to a list of tasks or features waiting to be addressed
16 this pattern injects dependencies into objects rather than having them construct internally
19 Use cases may often represent multiple ________
20 the one constant in software analysis and design

DOWN
1 time-boxed iteration in Scrum called
3 Objects in loosely coupled applications are more _____ than tightly coupled ones
4 refers to the process of combining different systems and components
5 Customers focus on this part of your applications
9 ________ requirements can lead to defects later in the development lifecycle
11 This kind of testing ensures new code doesn't break existing features

15 always code to this if possible
17 This process aims to unify software development and operations
18 Your applications should be easy to _______

Q2. (2 points)
Detail two major principles that impact the maintainability of a software system. What are the specific metrics used with respect to the two principles.

Q3. (2 points)
"Achieving complete test coverage is both feasible and should be a priority for complex software systems." Do you find this assertion valid? Support your answer with reasoning.

Q4. (2 points)
Your team is considering adopting continuous integration and continuous deployment (CI/CD) practices. What are the main benefits of implementing CI/CD pipelines in the software development lifecycle?

Q5. (2 points)
Discuss the difference between centralized version control and distributed version control. Give an example and reason for why one of preferred over another.

Q6. (2 points)
Explain the concept of Technical Debt in software engineering. Discuss how it can impact a software project in the long term and provide strategies for managing it effectively.

Q7. (2 points)
Describe the principles of Agile software development and how they contrast with traditional waterfall methodologies. Provide at least two advantages of Agile over waterfall.

Q8. (2 points)
What are the steps in MITRE ATT&CK framework?

Q9. (2 points)
List any 4 of Top 10 OWASP identified attacks?

Q10. (2 points)
Identify the following types of integration strategies:

(a) Start by testing only the user interface with the underlying functionality simulated by stubs:

(b) Testing the lower levels of the system via drivers:

Q11. (2 points)
You are working in a team of 10 people to build a math library product that contains over 150 mathematical operations, such as solving simultaneous equations, compute Fourier and Laplace transforms, solving differential equations, perform numerical integration etc. Each member of the team is working on a few of these routines. There is some common code used by all the routines to perform utility functions such as sorting values, and a Façade class that provides the external interface to the library. What type of integration approach would you choose for this application, and why?

Q12. (4 points)

You are working on software that helps users check arrival/departure times for flights for a particular airline. One of your colleagues suggests the following interface design:

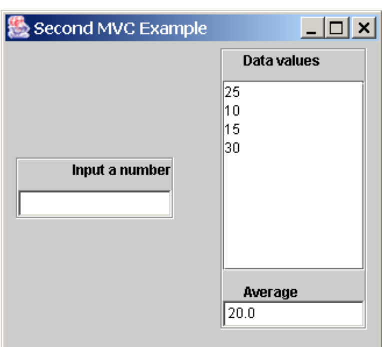
a) The initial screen presents a dialog box where users can type in the name of a departure/arrival city. If the city has multiple airports, or the name is not unique, it brings up a list of airport codes and asks users to select the airport code. If the city name is not found, it brings up a list of city names & airports serviced by the airline and asks the users to select from the list.

b) The next screen prompts users for a flight number. In case users don't know their flight number, there is a "Find flight number" button. If they enter an invalid flight number, a dialog box pops up: "Invalid flight number", with an "OK" button below it. Once users click on OK, they return to the flight number entry screen.

c) When a valid flight number is entered, the result shows up: the scheduled and actual arrival /departure times. The times are color-coded: red for AM, green for PM.

Identify 4 problems with this interface.

Q13. (4 points)



This is a screen shot from the running of a Java application. As the user enters numbers the program maintains an internal array to hold all the values entered by the user. After each number is entered the program adds it to the window of "Data values" and updates the "Average". Assume that the designer has associated a class named *Model* with the storage of the values in the array. In addition, there is a *Control* class to get new numbers and two classes, *View1* and *View2*, to take care of displaying the values and average, respectively. Explain which pattern can be applied to the design of this application. Your explanation should describe what happens when the application first starts running and each time that a new number is entered.

Q14. (4 points)
You are working on a calendar management system, that keeps a list of appointments with the date and time they occurred. You find that the current design has the following two modules:

DayView

a) Keep track of the list of appointments
    - Includes the classes Appointment and AppointmentList

b) Write appointments to file

c) Search the list of appointments and find the appointments for a given day

d) Display the list of appointments for that day

e) Edit configuration parameters, such as day start time and end time

f) Provide for adding, deleting and editing appointments
    - Calls the editing functionality in WeeklyView

WeeklyView

- Display appointments for a specified week
    - Calls DayView to retrieve appointments for each day
- Provides for adding, deleting and editing appointments
    - Uses the appointment list in DayView
- Edit configuration parameters, such as first and last day of week

Identify four principles of good design that are violated by the above design. For each, explain or give an example of how the design violates the principle

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Q15. (8 points)
Inspect the following code and identify any potential issues, inefficiencies, or areas for improvement.

```
def compute_factorial(n):
    if n < 0:
        raise ValueError("Input must be a non-negative integer.")
    elif n == 0:
        return 1
    else:
        return n * compute_factorial(n - 1)

def is_even(num):
    return num % 2 == 0

def generate_even_factorials(limit):
    even_factorials = []
    for i in range(limit):
        factorial_value = compute_factorial(i)
        if is_even(factorial_value):
            even_factorials.append(factorial_value)
    return even_factorials

      print(generate_even_factorials(10))
```

While typical inspection does not require you to suggest a solution, in this case, for each issue, inefficiency, improvement that you have identified, you are expected to suggest the fix and rewrite the specific lines of code.

Q16. (10 points) Home Automation System

You are tasked with designing a state diagram for a comprehensive Home Automation System that manages various functions within a residential environment. This system should accommodate a range of functionalities that include controlling lighting, heating, security, and entertainment devices based on user interactions and environmental conditions.

Develop a UML state diagram ensuring it captures the essential states and transitions based on the following specifications:

1. The system should have distinct operational modes that vary based on user preferences, time of day, or detected conditions (e.g., occupancy).
2. Each operational mode must dictate specific behaviors for the control of devices, including their activation and deactivation.
3. The system must be capable of responding to unexpected events, such as malfunctions or security breaches, and indicate the necessary responses in these scenarios.
4. Users should be able to switch between different modes manually or automatically based on scheduled actions or environmental triggers, prompting the system to react appropriately.
5. The system should distinguish between normal operational states and special states that require heightened measures or alerts, particularly in cases of emergencies.

Clearly list any/all assumptions you make while modeling the system.

Q17. (Three parts, 10 points total)

Automated Coffee Shop Ordering System

Develop an automated ordering system for the The Most Amazing Coffee Shop, where customers can place orders at self-service kiosks (SSKs) located inside the shop. Each order will include the customer's selection of drinks (e.g., espresso, latte, cappuccino), size (small, medium, large), any added customizations (e.g., milk type, flavor shots), and payment information. Once an order is placed and payment is confirmed, the system will generate a receipt, and the customer will be notified when their order is ready for pickup. The coffee shop staff will also be able to update the menu and availability of items in real-time.

Note: You may make simple assumptions about this problem based on your own knowledge of coffee shop ordering systems.

a) **(3 points)** Write a detailed textual use case (step-by-step) for a customer successfully placing an order using the SSK.

b) **(4 points)** Draw a UML Class diagram for the ordering system, showing the classes, relationships, and multiplicities between classes (e.g., Customer, Order, MenuItem, Payment).

c) **(3 points)** Based on your class diagram, create a UML sequence diagram for a customer successfully placing an order at the SSK.